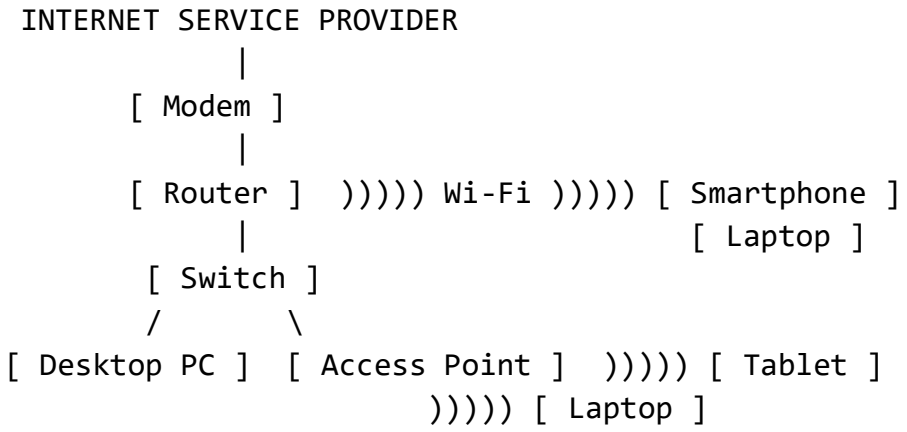
UNDERSTANDING AND MAINTENANCE OF NETWORK

Day 1: Introduction to Networking Hardware

1. Diagram of a Typical Home Wi-Fi Setup

The diagram below shows how internet reaches different devices in a home. The modem receives the internet connection from the service provider. The router shares that connection. A switch adds more wired ports, and an access point gives Wi-Fi coverage in another area.



Note: In many homes, the modem and router are combined in one device. The switch and extra access point are optional; they are useful when more wired ports or wider Wi-Fi coverage are needed.

2. Main Functions and Real-Life Examples

Device	Main function	Real-life example
Modem	A modem connects the home network to the internet service provider. It changes the signal so internet data can travel between the provider and the home.	The modem installed by a broadband company near the TV or computer.
Router	A router connects different networks and sends data to the correct device. It also shares one internet connection with many devices.	A Wi-Fi router used at home to connect phones, laptops and smart TVs.
Switch	A switch connects many wired devices in the same local network. It sends data to the device that needs it.	A switch in an office connecting desktop computers and printers.
Access Point (AP)	An access point provides Wi-Fi and allows wireless devices to join a wired network.	An access point in a school corridor to improve Wi-Fi coverage.
Network Cable	A network cable carries data between network devices. It can provide a stable wired connection.	An Ethernet cable connecting a desktop computer to a router.

3. Three Types of Network Cables

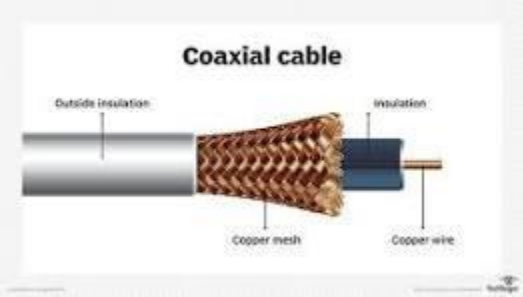
A. Ethernet Cable (Twisted Pair)

It is commonly used to connect computers, routers, switches and other devices in homes and offices.



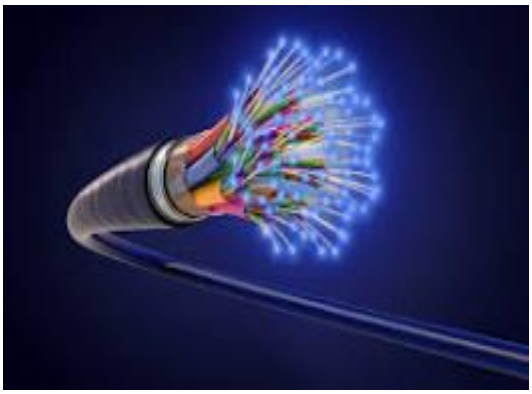
B. Coaxial Cable

It is commonly used for cable TV and some broadband internet connections.



C. Fiber Optic Cable

It carries data as light and is commonly used for high-speed internet and long-distance communication.



4. Difference Between a Router and a Switch

Point of difference	Router	Switch
Main use	Connects different networks and shares internet.	Connects devices inside the same local network.
Typical location	Usually placed near the modem or internet connection at home or office.	Usually placed in an office, server room or network cabinet.
How it handles data	Uses IP addresses to decide where data should go.	Uses MAC addresses to send data to the correct device on the local network.
Internet connection	Often connects the local network to the internet.	Does not normally connect a network to the internet by itself.
Wireless support	Many home routers also provide Wi-Fi.	A normal switch provides wired connections, not Wi-Fi.
Example	A home Wi-Fi router connecting phones and laptops to the internet.	An office switch connecting many computers and printers.